n-Dimensional Regional Sampling

1 Setup Hoeffding and Bernstein Approach

We have a function $f:\mathbb{R}^p\to\mathbb{R}$ that is differentiable and has a bounded first derivative, and we want to estimate $\int_{[0,1]^n} f(\boldsymbol{x})d\boldsymbol{x}$.

1. We write vector as $\boldsymbol{x}=(x^{(1)},\dots,x^{(p)})$.
2. We split $[0,1]^p$ as follows:

$$[0,1]^p=\bigcup_{i_1=0}^{m_1-1}\cdots\bigcup_{i_p=0}^{m_p-1}\left\{\left(\prod_{k=1}^{p-1}\left[a_{i_k}^{(k)},\,a_{i_k+1}^{(k)}\right)\right)\times\left[a_{i_p}^{(p)},\,a_{i_p+1}^{(p)}\right]\right\}.$$

   where for $1\le i\le p$, coordinate $a^{(i)}$ is split into m_i regions. We note $0=a_0^{(k)}<a_1^{(k)}<\cdots<a_{m_k}^{(k)}=1$ for all $1\le k\le p$. We denote the region

3. Denote total budget as N, and for region $\mathcal{A}[i_1,\dots,i_p]:=[a_{i_1}^{(1)},a_{i_1+1}^{(1)}]\times\cdots\times[a_{i_p}^{(p)},a_{i_p+1}^{(p)}]$ with $0\le i_j\le m_j-1$ for $1\le j\le p$, we sample $N[i_1,\dots,i_p]$, hence we have

$$N=\sum_{i_p=0}^{m_p-1}\cdots\sum_{i_1=0}^{m_1-1}N[i_1,\dots,i_p]$$

4. We estimate

$$\begin{aligned}\int_{\mathcal{A}[i_1,\dots,i_p]}f(\boldsymbol{x})d\boldsymbol{x}&=\int_{\mathcal{A}[i_1,\dots,i_p]}f(\boldsymbol{x})\cdot\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])\frac{1}{\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])}d\boldsymbol{x}\\&=\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])\cdot\mathbb{E}_{\boldsymbol{X}\sim\mathcal{U}(\mathcal{A}[i_1,\dots,i_p])}[f(\boldsymbol{X})]\\&\approx\frac{\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])}{N[i_1,\dots,i_p]}\sum_{k=1}^{N[i_1,\dots,i_p]}f(\boldsymbol{X}_k)\qquad\boldsymbol{X}_k\sim\mathcal{U}(\mathcal{A}[i_1,\dots,i_p])\end{aligned}$$

   with $0\le i_j\le m_j-1$ for $1\le j\le p$. Therefore, we get

$$\begin{aligned}I:=\int_{[0,1]^p}f(\boldsymbol{x})d\boldsymbol{x}&=\sum_{i_p=0}^{m_p-1}\cdots\sum_{i_1=0}^{m_1-1}\int_{\mathcal{A}[i_1,\dots,i_p]}f(\boldsymbol{x})d\boldsymbol{x}\\&=\sum_{i_p=0}^{m_p-1}\cdots\sum_{i_1=0}^{m_1-1}\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])\cdot\mathbb{E}_{\boldsymbol{X}\sim\mathcal{U}(\mathcal{A}[i_1,\dots,i_p])}[f(\boldsymbol{X})]\\&\approx\sum_{i_p=0}^{m_p-1}\cdots\sum_{i_1=0}^{m_1-1}\frac{\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])}{N[i_1,\dots,i_p]}\sum_{k=1}^{N[i_1,\dots,i_p]}f(\boldsymbol{X}_k\sim\mathcal{U}(\mathcal{A}[i_1,\dots,i_p]))\\&=\sum_{i_1,\dots,i_p}\sum_{k=1}^{N[i_1,\dots,i_p]}\frac{\mathrm{Vol}(\mathcal{A}[i_1,\dots,i_p])}{N[i_1,\dots,i_p]}f(\boldsymbol{X}_k\sim\mathcal{U}(\mathcal{A}[i_1,\dots,i_p]))=\hat{I}\end{aligned}$$

5. We have the following concentration bound

$$
\begin{aligned}
&\mathbb{P}(|\hat{I} - I| > \varepsilon) \\
&= \mathbb{P}\left(\left|\sum_{i_1,\ldots,i_p}\sum_{k=1}^{N[i_1,\ldots,i_p]} \frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])}{N[i_1,\ldots,i_p]}\left\{f(\boldsymbol{X}_i \sim \mathcal{U}(\mathcal{A}[i_1,\ldots,i_p])) - \mathbb{E}_{\boldsymbol{X}\sim\mathcal{U}(\mathcal{A}[i_1,\ldots,i_p])}[f(\boldsymbol{X})]\right\}\right| > \varepsilon\right)
\end{aligned}
$$

If we denote $Z_k[i_1,\ldots,i_p] = \frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])}{N[i_1,\ldots,i_p]} f(\boldsymbol{X}_k \sim \mathcal{U}(\mathcal{A}[i_1,\ldots,i_p]))$, then we have

$$
\mathbb{P}(|\hat{I} - I| > \varepsilon) = \mathbb{P}\left(\left|\sum_{i_1,\ldots,i_p}\sum_{k=1}^{N[i_1,\ldots,i_p]} \{Z_k[i_1,\ldots,i_p] - \mathbb{E}[Z_k[i_1,\ldots,i_p]]\}\right| > \varepsilon\right) \tag{1}
$$

for any $0 \le i_j \le m_j - 1, 1 \le j \le p$ we have $Z_k[i_1,\ldots,i_p]$ is bounded, and hence subgaussian.

1.1 Hoeffding approach

1.1.1 General Hoeffding

Now we apply Hoeffding to (1) to obtain

$$
\begin{aligned}
\mathbb{P}(|\hat{I} - I| > \varepsilon) &= \mathbb{P}\left(\left|\sum_{i_1,\ldots,i_p}\sum_{k=1}^{N[i_1,\ldots,i_p]} \{Z_k[i_1,\ldots,i_p] - \mathbb{E}[Z_k[i_1,\ldots,i_p]]\}\right| > \varepsilon\right) &(2)\\
&\le 2\exp\left(-\frac{c\varepsilon^2}{\sum\limits_{i_1,\ldots,i_p}\sum\limits_{k=1}^{N[i_1,\ldots,i_p]} \|Z_k[i_1,\ldots,i_p] - \mathbb{E}[Z_k[i_1,\ldots,i_p]]\|_{\psi_2}^2}\right) &(3)
\end{aligned}
$$

From the centering lemma, we have

$$
\begin{aligned}
\|Z_k[i_1,\ldots,i_p] - \mathbb{E}[Z_k[i_1,\ldots,i_p]]\|_{\psi_2}^2 &\le C\|Z_k[i_1,\ldots,i_p]\|_{\psi_2}^2 \\
&= C\left(\frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])}{N[i_1,\ldots,i_p]}\right)^2 \|f(\boldsymbol{X}_k \sim \mathcal{U}(\boldsymbol{x}[i_1,\ldots,i_p]))\|_{\psi_2}^2
\end{aligned}
$$

for some absolute constant C. From now on, we abbreviate $\sim \mathcal{U}(\mathcal{A}[i_1,\ldots,i_p])$ to just $\sim \mathcal{A}[i_1,\ldots,i_p]$. Substituting into (2) we obtain

$$
\begin{aligned}
\mathbb{P}(|\hat{I} - I| \ge \varepsilon) &\le 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum\limits_{i_1,\ldots,i_p}\sum\limits_{k=1}^{N[i_1,\ldots,i_p]} \left(\frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])}{N[i_1,\ldots,i_p]}\right)^2 \|f(\boldsymbol{X}_k \sim \mathcal{A}[i_1,\ldots,i_p])\|_{\psi_2}^2}\right) &(4)\\
&= 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum\limits_{i_1,\ldots,i_p} \frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])^2}{N[i_1,\ldots,i_p]} \|f(\boldsymbol{X} \sim \mathcal{A}[i_1,\ldots,i_p])\|_{\psi_2}^2}\right) &(5)
\end{aligned}
$$

where $\tilde{c} := c/C$ is absolute constant.

1.1.2 Hoeffding with max-min

We denote the min and max that f takes on $\mathcal{A}[i_1,\ldots,i_p]$ as $m[i_1,\ldots,i_p], M[i_1,\ldots,i_p]$ respectively, then we obtain another classic Hoeffding concentration bound from (2)

$$\mathbb{P}(|\hat{I} - I| \geq \varepsilon) \leq 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum\limits_{i_1,\ldots,i_p} \frac{\mathrm{Vol}(\mathcal{A}[i_1,\ldots,i_p])^2}{N[i_1,\ldots,i_p]} (M[i_1,\ldots,i_p] - m[i_1,\ldots,i_p])^2} \right) \tag{6}$$

1.1.3 Algorithm

Starting with $[0,1]^p$, before any splitting, all coordinates have a single (whole) region $m_1 = \cdots = m_p = 1$ and for each coordinate ℓ there are only two endpoint index $i_\ell \in \{0,1\}$ where $a_0^{(\ell)} = 0$ and $a_1^{(\ell)} = 1$ for $1 \leq \ell \leq p$.

We will split one coordinate per iteration of the algorithm. Suppose we split at coordinate j, that $m_j = 2$ and $m_k = 1$ for $k \neq j$. Further, assume that we split the j-coordinate at $t \in (0,1)$, that is $a_0^{(j)} = 0, a_1^{(j)} = t$ and $a_2^{(j)} = 1$, then we denote the region

$$\mathcal{A}_j^- = \mathcal{A}[0,\ldots,\underset{j\text{-th}}{0},\ldots,0], \qquad \mathcal{A}_j^+ = \mathcal{A}[0,\ldots,\underset{j\text{-th}}{t},\ldots,0]$$

and we allocate n^+ to region A_j^+ and $n^- = N - n^+$ to $\mathcal{A}_j^-$, thus we have

$$C_j(t) = \frac{\mathrm{Vol}(\mathcal{A}_j^-)^2}{n^-}(M_j^- - m_j^-)^2 + \frac{\mathrm{Vol}(\mathcal{A}_j^+)}{n^+}(M_j^+ - m_j^+)^2$$

thus we obtain the optimal splitting point t of j-th corodinate as

$$t_j^* = \arg\min_t C_j(t)$$

Correspondingly, we pick coordinate j^* for splitting as

$$j^* = \arg\min_j t_j^*$$

Then we repeat the above by replacing the interval $[0,1]^p$ with $\mathcal{A}_j^+$ and $\mathcal{A}_j^-$ respectively, and stopping criteria is either given by a stopping depth d or limited by number of samples per region.